

Friday, January 17

Using the `contrast` Function

Frequently a quantity of interest has the form

$$E(Y_a) - E(Y_b),$$

where $E(Y_a)$ and $E(Y_b)$ represent the expected value of the response variable under circumstances a and b , respectively. By “circumstances” we mean when explanatory variable(s) assume(s) specific values. This is sometimes called a “contrast” and inferences concerning a contrast can be made using the `contrast` function from the `trtools` package.¹ For linear models, any contrast can be expressed as a linear function of the model parameters, and so `contrast` can be used instead of `lincon` if desired.

Example: Consider a linear model for the `anorexia` data.

```
library(MASS) # for anorexia data frame
anorexia$change <- anorexia$Postwt - anorexia$Prewt # compute weight change
head(anorexia) # inspect "head" of anorexia data frame
```

```
  Treat Prewt Postwt change
1  Cont  80.7   80.2   -0.5
2  Cont  89.4   80.1   -9.3
3  Cont  91.8   86.4   -5.4
4  Cont  74.0   86.3   12.3
5  Cont  78.1   76.1   -2.0
6  Cont  88.3   78.1  -10.2
```

```
summary(anorexia) # summary of variables in anorexia data frame
```

	Treat	Prewt	Postwt	change
CBT :29	Min.	:70.0	Min. : 71.3	Min. : -12.20
Cont:26	1st Qu.:	:79.6	1st Qu.: 79.3	1st Qu.: -2.23
FT :17	Median :	:82.3	Median : 84.0	Median : 1.65
	Mean :	:82.4	Mean : 85.2	Mean : 2.76
	3rd Qu.:	:86.0	3rd Qu.: 91.5	3rd Qu.: 9.10
	Max. :	:94.9	Max. :103.6	Max. : 21.50

```
str(anorexia) # structure of anorexia data frame
```

```
'data.frame': 72 obs. of 4 variables:
 $ Treat : Factor w/ 3 levels "CBT","Cont","FT": 2 2 2 2 2 2 2 2 2 2 ...
 $ Prewt : num 80.7 89.4 91.8 74 78.1 88.3 87.3 75.1 80.6 78.4 ...
 $ Postwt: num 80.2 80.1 86.4 86.3 76.1 78.1 75.1 86.7 73.5 84.6 ...
 $ change: num -0.5 -9.3 -5.4 12.3 -2 ...
```

```
m <- lm(change ~ Treat, data = anorexia)
summary(m)$coefficients
```

¹The `contrast` function in the `trtools` package is modeled after functions of the same name in the `contrast` and `rms` packages. Mine has some features not found in those functions, but I did not incorporate all of the features in those functions either. You will also learn how to use functions from the `emmeans` package which have some functionality like that of `contrast` from the `trtools` package.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.007	1.398	2.151	0.03499
TreatCont	-3.457	2.033	-1.700	0.09361
TreatFT	4.258	2.300	1.852	0.06838

We can see that the model is

$$E(Y_i) = \begin{cases} \beta_0, & \text{if the treatment for the } i\text{-th observation is cognitive-behavioral therapy,} \\ \beta_0 + \beta_1, & \text{if the treatment for the } i\text{-th observation is control,} \\ \beta_0 + \beta_2, & \text{if the treatment for the } i\text{-th observation is family therapy.} \end{cases}$$

Suppose we want to make inferences about the difference in the expected weight change between the family therapy and the control conditions. We have that

$$\begin{aligned} E(Y_a) &= \beta_0 + \beta_1, \\ E(Y_b) &= \beta_0 + \beta_2, \end{aligned}$$

and so this difference is

$$E(Y_a) - E(Y_b) = \underbrace{\beta_0 + \beta_2}_{E(Y_a)} - \underbrace{(\beta_0 + \beta_1)}_{E(Y_b)} = \beta_2 - \beta_1.$$

Inferences concerning $\beta_2 - \beta_1$ can be made using `lincon` since this is a linear function of the model parameters:

$$\ell = 0 \times \beta_0 + (-1) \times \beta_1 + 1 \times \beta_2 = \beta_2 - \beta_1.$$

```
library(trtools) # loading for the lincon and contrast functions
lincon(m, a = c(0,-1,1))
```

```
      estimate      se lower upper tvalue df   pvalue
(0,-1,1),0    7.715 2.348  3.03  12.4  3.285 69 0.001602
```

Alternatively we can use the `contrast` function in which we specify values of the explanatory variables rather than having work out the coefficients of the linear function.

```
contrast(m, a = list(Treat = "FT"), b = list(Treat = "Cont"))
```

```
      estimate      se lower upper tvalue df   pvalue
      7.715 2.348  3.03  12.4  3.285 69 0.001602
```

Note: The meaning of the `a` and `b` arguments are different for the `lincon` and `contrast` functions.

The `contrast` function can compute multiple contrasts at a time. Suppose we want to estimate the difference in the expected weight change between each of the two therapy conditions and the control condition:

```
contrast(m, a = list(Treat = c("CBT","FT")), b = list(Treat = "Cont"))
```

```
      estimate      se lower upper tvalue df   pvalue
      3.457 2.033 -0.5994  7.513  1.700 69 0.093608
      7.715 2.348  3.0302 12.399  3.285 69 0.001602
```

Note that `contrast` includes an optional `cnames` (contrast names) argument to label the output for clarity.

```
contrast(m,
  a = list(Treat = c("CBT","FT")),
  b = list(Treat = "Cont"),
  cnames = c("CBT vs Control","FT vs Control"))
```

```
      estimate      se lower upper tvalue df   pvalue
CBT vs Control  3.457 2.033 -0.5994  7.513  1.700 69 0.093608
FT vs Control   7.715 2.348  3.0302 12.399  3.285 69 0.001602
```

Warning: To avoid confusion, do not list multiple values of more than one explanatory variable. It is possible to specify multiple values of two or more explanatory variables, but the output can be confusing.

If we do not specify the `b` argument then `contrast` will produce an estimate of just $E(Y_a)$. For example, we can estimate the expected weight change under each condition.

```
contrast(m, a = list(Treat = c("CBT", "FT", "Cont")),
  cnames = c("Cognitive-Behavioral", "Family", "Control"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Cognitive-Behavioral	3.007	1.398	0.218	5.796	2.1509	69	0.0349920
Family	7.265	1.826	3.622	10.907	3.9787	69	0.0001688
Control	-0.450	1.476	-3.395	2.495	-0.3048	69	0.7614470

Example: Consider again a model for the `whiteside` data.

```
m <- lm(Gas ~ Insul + Temp + Insul:Temp, data = whiteside)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.8538	0.13596	50.409	7.997e-46
InsulAfter	-2.1300	0.18009	-11.827	2.316e-16
Temp	-0.3932	0.02249	-17.487	1.976e-23
InsulAfter:Temp	0.1153	0.03211	3.591	7.307e-04

Recall that the model can be written as

$$E(G_i) = \begin{cases} \beta_0 + \beta_2 t_i, & \text{if the } i\text{-th observation is before insulation,} \\ \beta_0 + \beta_1 + (\beta_2 + \beta_3) t_i, & \text{if the } i\text{-th observation is after insulation.} \end{cases}$$

So the rate of change in expected gas consumption per unit increase in temperature is β_2 before insulation and $\beta_2 + \beta_3$ after insulation. Inferences concerning both of these quantities can be obtained using `lincon`.

```
lincon(m, a = c(0,0,1,0)) # b2 (note: also shown by summary above)
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,1,0),0	-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23

```
lincon(m, a = c(0,0,1,1)) # b2 + b3
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,1,1),0	-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

To use `contrast` to produce these inferences we consider a one unit change temperature.

```
contrast(m,
  a = list(Insul = "Before", Temp = 1),
  b = list(Insul = "Before", Temp = 0))
```

	estimate	se	lower	upper	tvalue	df	pvalue
	-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23

```
contrast(m,
  a = list(Insul = "After", Temp = 1),
  b = list(Insul = "After", Temp = 0))
```

	estimate	se	lower	upper	tvalue	df	pvalue
	-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

```
contrast(m,
  a = list(Insul = c("Before", "After"), Temp = 1),
```

```
b = list(Insul = c("Before","After"), Temp = 0),
cnames = c("before","after"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
before	-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23
after	-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

Note that since the rate of change in expected gas consumption is *constant*, any two values of temperature one unit apart would produce the same result. For example:

```
contrast(m,
a = list(Insul = c("Before","After"), Temp = 3),
b = list(Insul = c("Before","After"), Temp = 2),
cnames = c("before","after"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
before	-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23
after	-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

What is the change in expected gas consumption if temperature goes up by 10C?

```
contrast(m,
a = list(Insul = c("Before","After"), Temp = 12),
b = list(Insul = c("Before","After"), Temp = 2),
cnames = c("before","after"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
before	-3.932	0.2249	-4.384	-3.481	-17.49	52	1.976e-23
after	-2.779	0.2292	-3.239	-2.319	-12.12	52	8.936e-17

Expected gas consumption at 5C before and after insulation are $\beta_0 + \beta_2 5$ and $\beta_0 + \beta_1 + (\beta_2 + \beta_3)5$. Inferences concerning these quantities can be obtained as follows.

```
lincon(m, a = c(1,0,5,0)) # b0 + b2 x 5
```

	estimate	se	lower	upper	tvalue	df	pvalue
(1,0,5,0),0	4.888	0.06383	4.76	5.016	76.57	52	3.885e-55

```
lincon(m, a = c(1,1,5,5)) # b0 + b1 + (b2 + b3)5
```

	estimate	se	lower	upper	tvalue	df	pvalue
(1,1,5,5),0	3.334	0.06024	3.213	3.455	55.35	52	6.772e-48

```
contrast(m,
a = list(Insul = c("Before","After"), Temp = 5),
cnames = c("before @ 5C","after @ 5C"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
before @ 5C	4.888	0.06383	4.760	5.016	76.57	52	3.885e-55
after @ 5C	3.334	0.06024	3.213	3.455	55.35	52	6.772e-48

The difference in expected gas consumption between before and after insulation at 5C is $\beta_1 + \beta_3 5$. Inferences concerning this quantity can be obtained as follows.

```
lincon(m, a = c(0,1,0,5))
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,1,0,5),0	-1.553	0.08777	-1.73	-1.377	-17.7	52	1.155e-23

```
contrast(m,
  a = list(Insul = "After", Temp = 5),
  b = list(Insul = "Before", Temp = 5))
```

```
estimate      se lower upper tvalue df      pvalue
-1.553 0.08777 -1.73 -1.377 -17.7 52 1.155e-23
```

In many cases we can use either `lincon` or `contrast`. The latter is often easier to use since it does not require the user to work out the coefficients for the linear function of model parameters. But there are cases where `lincon` can do something that `contrast` cannot.

Computing and Plotting Estimated Expected Responses

Recall that we can use `contrast` to compute estimated expected responses at specified values of the explanatory variables.

```
contrast(m, a = list(Insul = c("Before","After"), Temp = -1),
  cnames = c("before @ -1","after @ -1"))
```

```
estimate      se lower upper tvalue df      pvalue
before @ -1    7.247 0.1562 6.934 7.561 46.39 52 5.473e-44
after @ -1     5.002 0.1384 4.724 5.280 36.13 52 1.640e-38
```

But to create several estimates of the expected response it is more convenient to use the `predict` function (so-called because an estimate of $E(Y)$ can be also be viewed as a *prediction* of the value of Y). First let's create a data frame where we'd like to obtain estimates of the expected response.

```
d <- expand.grid(Insul = c("Before","After"), Temp = seq(-1, 11, by = 1))
head(d)
```

```
  Insul Temp
1 Before  -1
2 After   -1
3 Before   0
4 After    0
5 Before   1
6 After    1
```

```
tail(d)
```

```
  Insul Temp
21 Before   9
22 After    9
23 Before  10
24 After   10
25 Before  11
26 After   11
```

There are a couple of things to note here: the `expand.grid` function creates a data frame for *all combinations* of the values of the variables, and `seq` creates a *sequence* of values. For example,

```
seq(-1, 11, by = 2)      # sequence from -1 to 11 by increments of two
```

```
[1] -1  1  3  5  7  9 11
```

```
seq(-1, 11, length = 5) # sequence from -1 to 11 of five values
```

```
[1] -1  2  5  8 11
```

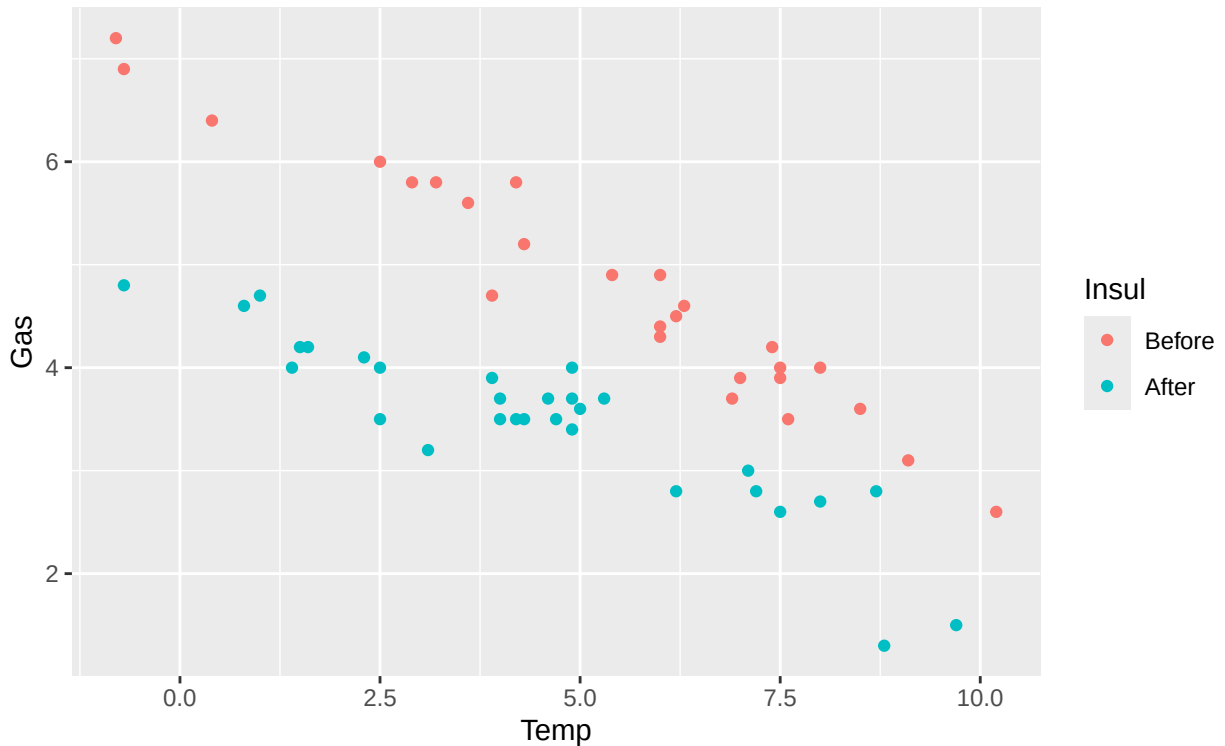
The `predict` function can be used to compute the estimated expected response for each pseudo-observation.

```
d$ey <- predict(m, newdata = d)
head(d)
```

```
   Insul Temp    ey
1 Before  -1 7.247
2 After   -1 5.002
3 Before   0 6.854
4 After    0 4.724
5 Before   1 6.461
6 After    1 4.446
```

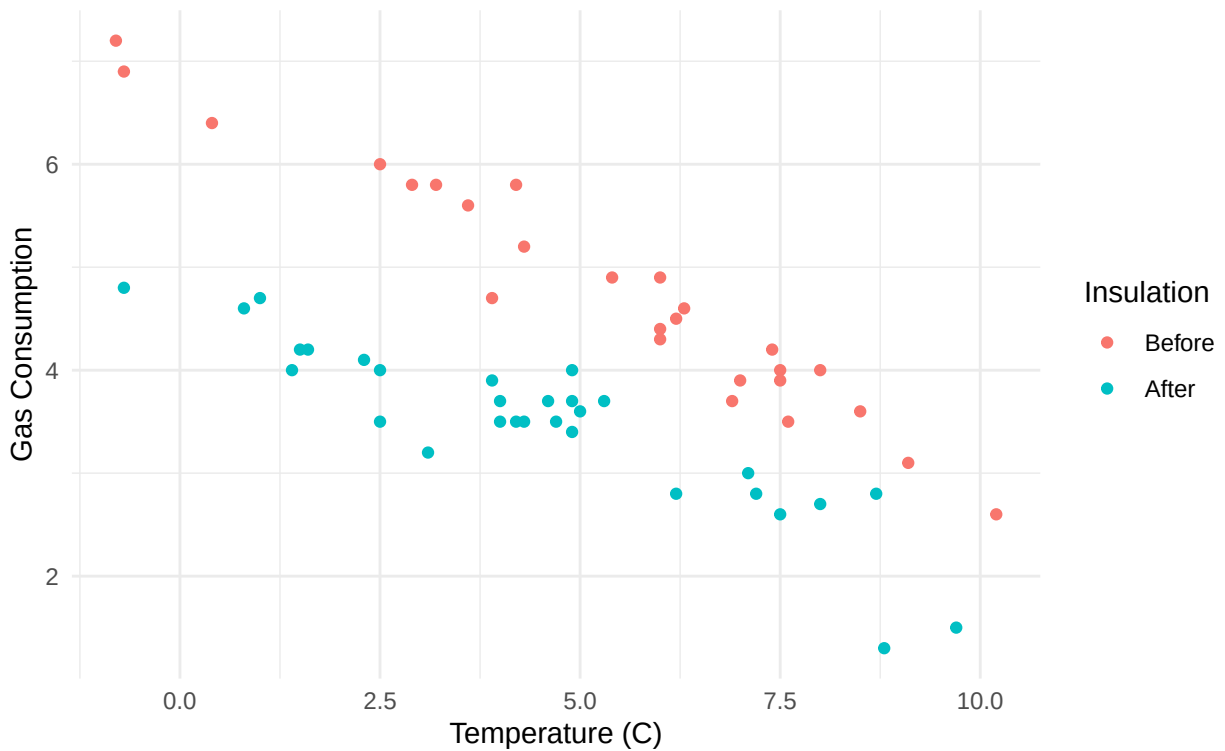
Now let's see if we can show expected gas consumption as a function of insulation and temperature. First we will plot the data.

```
library(ggplot2)
p <- ggplot(whiteside, aes(x = Temp, y = Gas, color = Insul)) + geom_point()
plot(p)
```



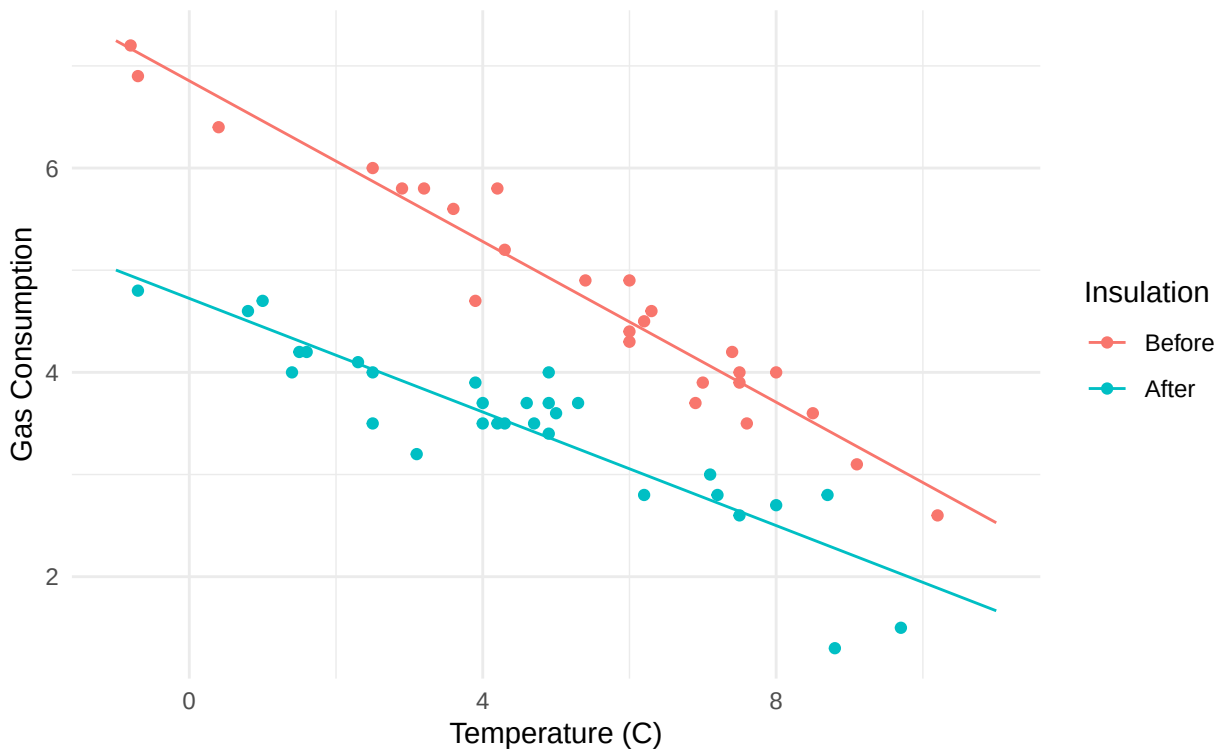
Let's change the aesthetic labels and the theme.

```
p <- ggplot(whiteside, aes(x = Temp, y = Gas, color = Insul)) + geom_point() +
  labs(x = "Temperature (C)", y = "Gas Consumption", color = "Insulation") +
  theme_minimal()
plot(p)
```



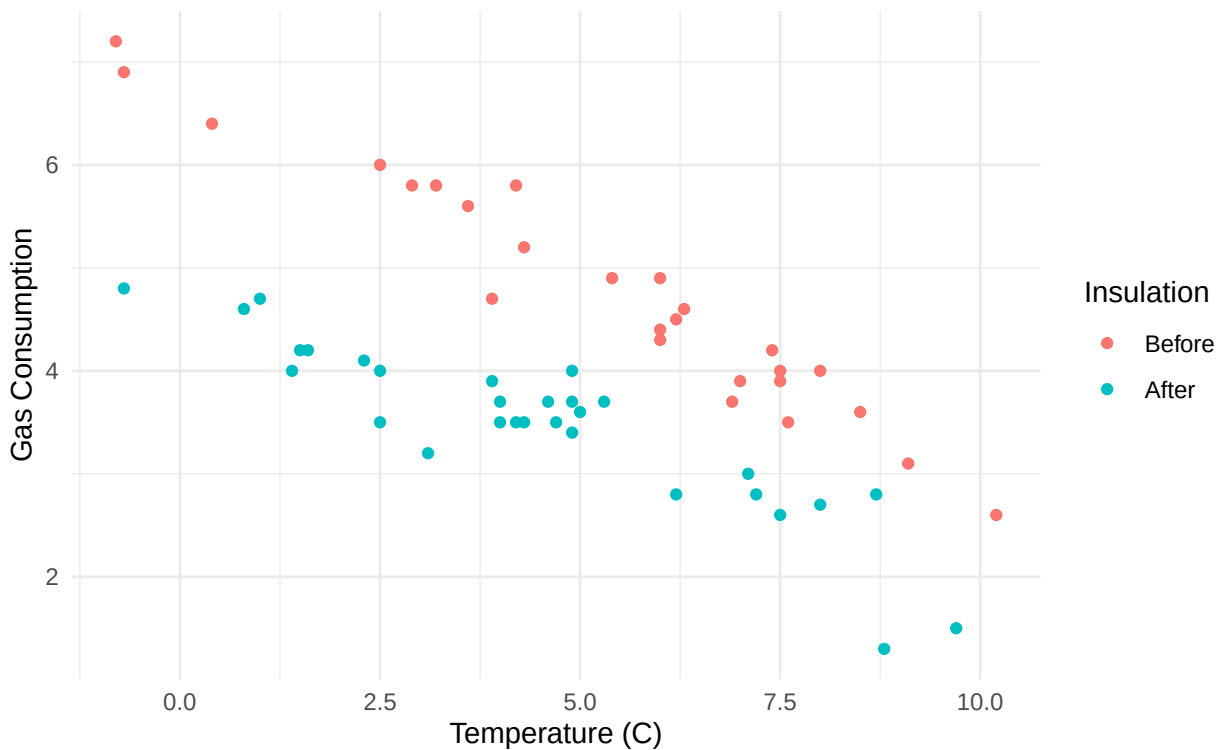
Now we can add lines to represent expected gas consumption as a function of temperature and insulation. Note that we need to change the y variable and the data frame for this part.

```
p <- ggplot(whiteside, aes(x = Temp, y = Gas, color = Insul)) + geom_point() +
  labs(x = "Temperature (C)", y = "Gas Consumption", color = "Insulation") +
  theme_minimal() + geom_line(aes(y = ey), data = d)
plot(p)
```

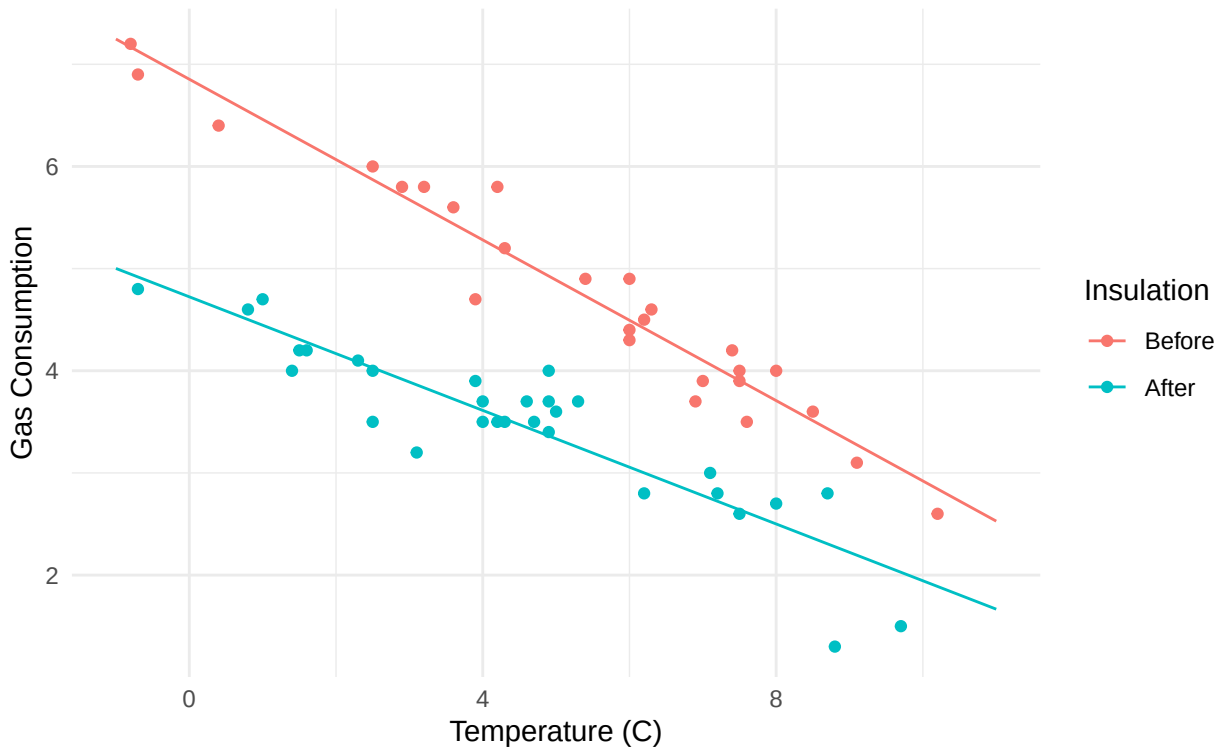


Note that you can do this in pieces, which might be easier to debug.

```
p <- ggplot(whiteside, aes(x = Temp, y = Gas, color = Insul)) + geom_point() +
  labs(x = "Temperature (C)", y = "Gas Consumption", color = "Insulation") +
  theme_minimal()
plot(p)
```




```
p <- p + geom_line(aes(y = ey), data = d)
plot(p)
```



Now consider the anorexia data and model from earlier.

```
m <- lm(change ~ Treat, data = anorexia)
contrast(m, a = list(Treat = c("CBT", "FT", "Cont")),
         cnames = c("Cognitive-Behavioral", "Family", "Control"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Cognitive-Behavioral	3.007	1.398	0.218	5.796	2.1509	69	0.0349920
Family	7.265	1.826	3.622	10.907	3.9787	69	0.0001688
Control	-0.450	1.476	-3.395	2.495	-0.3048	69	0.7614470

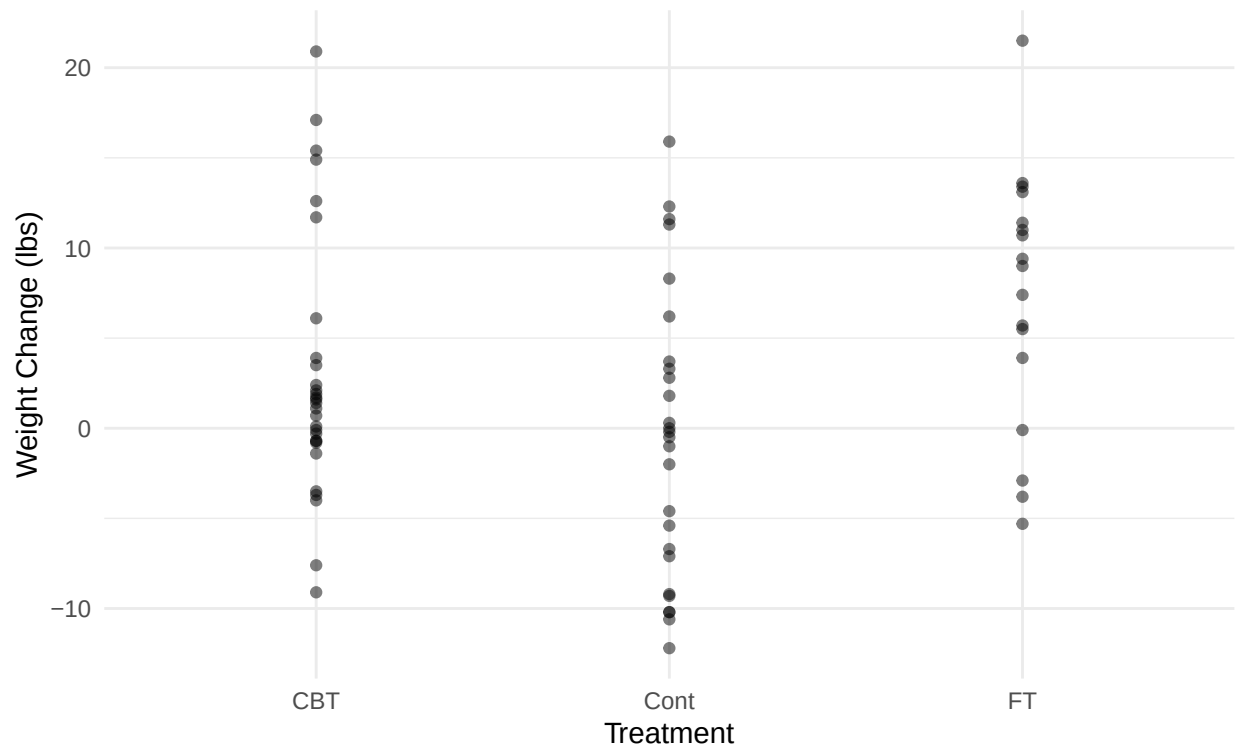
```
d <- data.frame(Treat = c("CBT", "FT", "Cont"))
d$yhat <- predict(m, newdata = d)
d
```

```
  Treat  yhat
1  CBT  3.007
2  FT   7.265
3  Cont -0.450
```

Note: We can use `data.frame` to create a data frame for generating estimated expected responses when there is just one explanatory variable. But if we want to create a data frame with various combinations of values of two or more explanatory variables it is easier to use `expand.grid`.

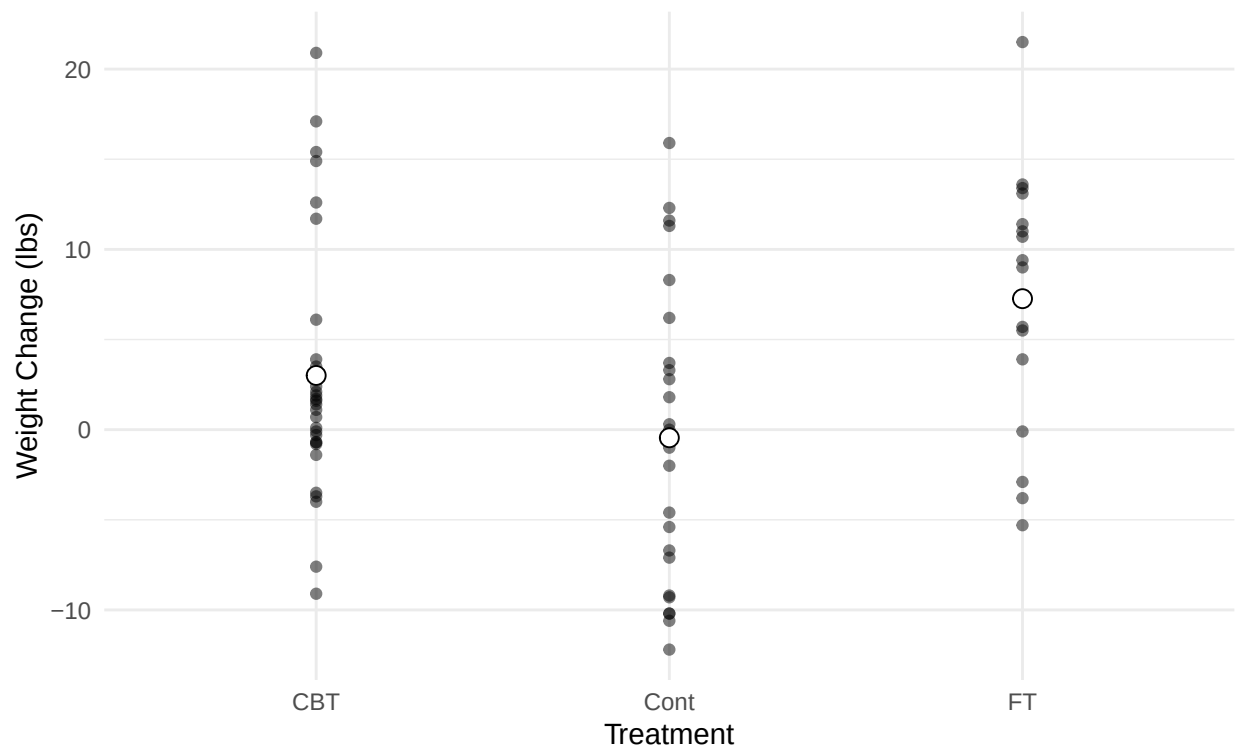
Here is a plot of the raw data.

```
p <- ggplot(anorexia, aes(x = Treat, y = change)) + theme_minimal() +
  geom_point(alpha = 0.5) + labs(x = "Treatment", y = "Weight Change (lbs)")
plot(p)
```



Note the use of `alpha = 0.5` to control the transparency of the points. Now we can add the estimated expected responses.

```
p <- p + geom_point(aes(y = yhat), data = d, shape = 21, size = 3, fill = "white")
plot(p)
```



Hint: Try a Google image search for “R point shapes” to know what number to use. Shape numbers 21-25 let

you separately specify the colors of the point fill and outline. For other aesthetics (e.g., size) you sometimes just have to experiment.

Maybe that would look better sideways?

```
p <- p + coord_flip()
plot(p)
```

